

1 **Projection-Based Safety Shielding for Provably Direction-Compliant COLREGs Collision Avoidance**
2 **in Continuous Action Spaces**

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1 ABSTRACT

2 **Objectives.** Autonomous surface vessels must obey the International Regulations for Preventing Collisions
3 at Sea (COLREGs). The give-way rules are directional. A track that avoids a collision but passes on
4 the wrong side is still a violation. This paper builds a per-step guarantee of directional compliance for
5 continuous control commands.

6 **Methods.** We write the give-way clause as a half-plane constraint on the plane of control commands.
7 We intersect it with the actuator limits and a one-step collision-free condition, which gives a convex safe set.
8 A small quadratic program projects the policy command onto the nearest point of that set. The projection
9 sits inside the environment transition, so the policy learns under its own corrected commands. The data are
10 2,000 simulated two-ship encounters, split into 1,400 training and 600 test cases. Nine configurations form
11 a single-variable ladder, each trained with eight fixed seeds.

12 **Findings.** On every give-way step where the projection is feasible, directional compliance holds by
13 construction. It does not depend on the training outcome or the random seed. Steps that switch to the
14 fallback branch are not covered. On the test set, the violation count per episode is [TBD] and the collision
15 rate is [TBD]. The ablation gains are significant under a paired signed-rank test. We do not claim the arrival
16 rate as a benefit.

17 **Novelty.** To our knowledge, this is the first projection-based safety mechanism that enforces directional
18 COLREGs compliance on continuous control commands. We also give a graded statement of what is proved
19 and of what is not.

20 **Practical Applications.** The compliance check is independent of training. The mechanism can sit above
21 an existing controller as a compliance layer for functional verification and liability assessment. The online
22 cost is one small quadratic program per step, so it runs in real time without special hardware.

23 **Keywords:** COLREGs compliance, safe reinforcement learning, projection-based safety shield, continuous
24 action space, quadratic programming, unmanned surface vehicle

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